

✓ 1 coll 2021
✓ 2 coll 2021
✓ 3 coll 2021
✓ 4 coll 2021

✓ 1 coll 2022
✓ 2 coll 2022
✓ 3 coll 2022
✓ 4 coll 2022

✓ 1 coll 2023
✓ 2 coll 2023

✓ HW 20/21
✓ HW 1 (kmms)
✓ HW 2 (kmms)
✓ HW 1,2

3° coll 2023 (KAT)

$$V_0 = 5V$$

$$V_G = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} = 3.207V$$

$$R_B = R_{B2} \parallel R_{B1} = 222.18 k\Omega$$

$$V_B - R_B I_B - V_{BE} - R_{E1} \left(I_C \left(1 + \frac{1}{\beta_F} \right) + I_{RDE} \right) = 0$$

$$\frac{V_{CC} - V_0}{R_{D2}} = I_D + I_{RDE}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$V_{GS} = V_{CC} - I_C R_{E1} - I_D R_{S2}$$

$$I_D = \frac{V_{CC} - V_{GS} - I_C R_{E1}}{R_{S2}}$$

$$I_{RDE} = \frac{V_0 - \left(V_B - \frac{I_C R_B}{\beta_F} - V_{BE} \right)}{R_{D2}}$$

$$\frac{V_B - R_B I_C}{\beta_F} - V_{BE} - R_{E1} \left[I_C \left(1 + \frac{1}{\beta_F} \right) + \frac{V_0 - \left(V_B - \frac{I_C R_B}{\beta_F} - V_{BE} \right)}{R_{D2}} \right] = 0$$

$$V_B - V_{BE} - \frac{R_{E1}}{R_{D2}} (V_0 - V_B + V_{BE}) = I_C \left[\frac{R_B}{\beta_F} + R_{E1} \left(1 + \frac{1}{\beta_F} \right) + \frac{R_{E1}}{R_{D2}} \left(\frac{R_B}{\beta_F} \right) \right]$$

$$I_C = 0.969 \text{ mA}$$

$$I_{RDE} = 0.516 \text{ mA}$$

$$I_D = 0.219 \text{ mA}$$

$$V_{GS} = 2.936 V$$

$$R_{S2} = \frac{V_{CC} - I_C R_{E1} - V_{GS}}{I_D} = 17.654 k\Omega$$

$$V_{CE} = V_{CC} - I_C R_{E1} - R_{E1} \left(I_C \left(1 + \frac{1}{\beta_F} \right) + I_{RDE} \right) = 5.312 V$$

$$V_{DS} = V_0 - I_D R_{S2} = 1.13 V$$

Voltage mixing voltage sampling

$$A_F = R_{DE} \parallel R_{E1} = 917 \Omega$$

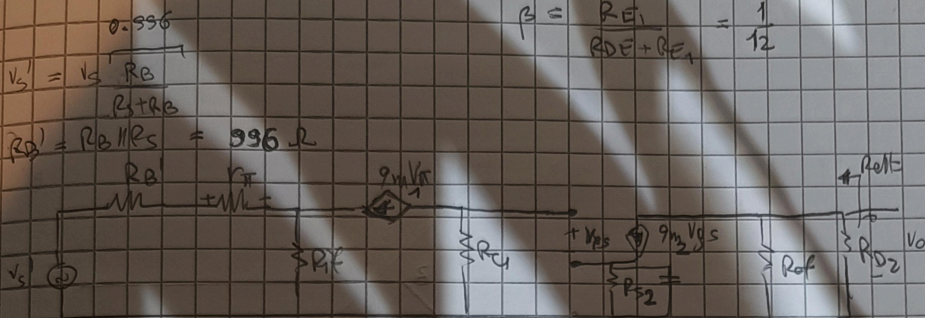
$$R_F = R_{DE} + R_{E1} = 12 \text{ k}\Omega$$

$$\beta = \frac{R_{E1}}{R_{DE} + R_{E1}} = \frac{1}{12}$$

$$g_{m1} = \frac{I_{C1}}{V_T} = 0.0323 \text{ A/V}$$

$$r_{\pi1} = 5.366 \text{ k}\Omega$$

$$g_{m2} = \frac{2I_0}{V_{GS} - V_{th}} = 4.68 \cdot 10^{-4} \text{ A/V}$$



$$A_d = \frac{N_0}{N_0} \Big|_{\beta=0}$$

$$A_d = \alpha_j \cdot \alpha_{CER} \cdot \alpha_{CS} = 7.027 \text{ [V/V]}$$

$$\alpha_{CS} = -g_{m2} R_{of} \parallel R_{D2} = -2.03 \text{ [V/V]}$$

$$A_F = \frac{A_d}{1 + \beta A_d} = \frac{7.027}{1 + 0.0833 \cdot 7.027} = 4.432 \text{ [V/V]}$$

$$\alpha_{CER} = \frac{-\beta_0 R_{C1}}{R_{inCER}} = -3.679 \text{ [V/V]}$$

$$A_V = \frac{V_o}{V_s} = A_F \cdot \frac{R_B}{R_s + R_B} = 4.414 \text{ [V/V]}$$

$$R_{inCER} = r_{\pi1} + (1 + \beta_0) R_F = 189.683 \text{ k}\Omega$$

$$\alpha_j = \frac{R_{inCER}}{R_{B1} + R_{inCER}} = 0.995 \text{ [V/V]}$$

$$R_{in}^F = R_{B1} + R_{inCER} = 190.679 \text{ k}\Omega$$

$$R_{in}^F = (R_{B1} + R_{inCER}) (1 + \beta A_d) = 302.36 \text{ k}\Omega$$

$$R_{out}^Q = R_{of} \parallel R_{D2} = 4.36 \text{ k}\Omega$$

$$R_{in} = (R_{in}^F - R_{B1}) \parallel R_B + R_s$$

$$= 128.89 \text{ k}\Omega$$

$$R_{out}^F = \frac{R_{out}^Q}{1 + \beta A_d} = 2.737 \text{ k}\Omega$$

$$R_{out} = R_{out}^F$$

C_s and C_{s2}

$$C_s: f_{LCS} = \frac{1}{2\pi C_s R_s^{SC}} = 12.35 \text{ Hz} \quad R_s^{SC} = R_s + (R_B \parallel (R_{in}^F - R_{B1})) = R_{in}$$

$$C_{s2}: R_{s2}^{SC} = R_{s2} \parallel \frac{1}{g_{m2}} = 1.906 \text{ k}\Omega$$

→ applying feedback

$$f_{Lc2} = \left(\frac{1}{2\pi C_{s2} R_{s2}^{SC}} \right) \frac{1}{1 + \beta A_d} = 5.275 \text{ Hz}$$

$$f_L = f_{LCS} + f_{Lc2} = 17.62 \text{ Hz}$$

Ex 2

$$1) A_v = \frac{V_o}{V_s} = 1 + \frac{R_2}{R_1} = 3.06$$

$$2) \beta_o = \frac{R_1}{R_1 + A_2}$$

$$\frac{1}{\beta_o} = 1 + \frac{R_2}{R_1} = 3.06 = 10 \text{ dB}$$

$$W_{CR} : \begin{cases} A_{db} W_o = A_v W \\ A_v W^2 = \frac{1}{\beta_o} W_{CR}^2 \end{cases}$$

$$W_{CR} = \sqrt{\beta_o W_o A_{db}}$$

$$= 2.56 \cdot 10^5 \text{ rad/s}$$

$$PS1 = 180 - \arctan\left(\frac{W_{CR}}{W_o}\right) - \arctan\left(\frac{W_{CR}}{W_1}\right) = 4.69^\circ \text{ au}$$

$$3) \beta_c = \beta_o \cdot \frac{1 + sC_c R_c}{1 + sC_c R_c + sC_c R_1 / R_2} = \beta_o \cdot \frac{1 + sC_c R_c}{1 + sC_c (R_c + R_1 / R_2)}$$

$$W_B = \frac{1}{C_c R_c} = 1.377 \cdot 10^3 \text{ rad/s}$$

$$W_p = \frac{1}{C_c (R_c + R_1 / R_2)} = 1.36 \text{ rad/s}$$

$$\frac{1}{\beta_o} = 1 + \frac{R_2}{R_1} = 10 \text{ dB}$$

$$\frac{1}{\beta_{oo}} = \left(1 + \frac{R_1}{R_2}\right) \left(\frac{R_c + R_1 / R_2}{R_c}\right) = 30.93 = 70 \text{ dB}$$

$$W_{CR}' = \beta_o W_o A_{db} = 3.233 \cdot 10^3 \text{ rad/s}$$

$$PS1 = 180 - \arctan\left(\frac{W_{CR}'}{W_o}\right) - \arctan\left(\frac{W_{CR}'}{W_1}\right) - \arctan\left(\frac{W_{CR}'}{W_p}\right) + \arctan\left(\frac{W_{CR}'}{W_B}\right) = 59.5^\circ$$

$$\text{Bandwidth} = W_{CR}' = 3.233 \cdot 10^3 \text{ rad/s}$$

$$4) A_v(s) = \frac{V_o}{V_s} = A_F = \frac{1}{\beta_c} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{1 + sC_c(R_c + R_1/R_2)}{1 + sC_c R_c}\right)$$

T >> 1
ideally A_v = ∞

